

Introduction to Cloud Computing

Cloud computing refers to the delivery of computing services, including servers, storage, databases, networking, and software, over the internet, allowing organizations and individuals to access resources on demand without needing to own and maintain physical infrastructure. This model offers significant advantages over traditional on-premises computing, including reduced upfront capital expenditure, since organizations can pay only for the resources they actually use rather than investing heavily in physical hardware that may sit underutilized. Cloud computing services are generally categorized into three primary models: infrastructure as a service, which provides fundamental computing resources such as virtual machines and storage; platform as a service, which provides a development and deployment environment, allowing developers to build applications without managing underlying infrastructure; and software as a service, which delivers complete software applications over the internet, accessible through a web browser without requiring local installation. Major cloud service providers, including Amazon Web Services, Microsoft Azure, and Google Cloud Platform, offer extensive ranges of services spanning these categories, competing on factors such as pricing, performance, geographic availability, and specialized service offerings. Scalability is one of the most significant benefits of cloud computing, allowing organizations to quickly increase or decrease their computing resources in response to changing demand, a capability particularly valuable for applications experiencing variable or unpredictable workloads. Cloud computing has fundamentally transformed how organizations approach information technology infrastructure, enabling faster innovation, improved disaster recovery capabilities, and global reach that would have been prohibitively expensive or complex to achieve through traditional on-premises infrastructure alone.